|| Jai Sri Gurudev ||
Sri Adichunchanagiri Shikshana Trust (R)
SJB Institute of Technology
BGS Health and Education City, Dr. Vishnuvardhana Road, Kengeri, Bengaluru-560060
Approved by AICTE, New Delhi.
Autonomous Institution affiliated to Visvesvaraya Technological University, Belagavi
Recognized by UGC, New Delhi with 2(f) & 12 (B), Accredited by NAAC with 'A+'grade,
Certified by ISO 9001 - 2015

A

Mini Project Report

On

"FocusAI: Real-Time Student Attention Tracking System"

SUBMITTED IN PARTIAL FULFILLMENT FOR THE AWARD OF MARKS OF

VI SEMESTER COURSE "AIML"

IN

Computer Science & Engineering

SUBMITTED BY

Vedashree K A (1JB24CS414)

Under the Guidance of

Mr. Santhosh K
Assistant Professor
Dept. of CSE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
No.67, BGS Health & Education City, Dr.Vishnuvardhan Rd, Kengeri, Bengaluru, Karnataka 560060

2025 – 2026

|| Jai Sri Gurudev ||
Sri Adichunchanagiri Shikshana Trust ®

SJB INSTITUTE OF TECHNOLOGY

No.67, BGS Health & Education City, Dr.Vishnuvardhan Rd, Kengeri, Bengaluru, Karnataka 560060

Department of Computer Science and Engineering

CERTIFICATE

Certified that the Mini Project/Formative Assessment entitled "**FocusAI: Real-Time Student Attention Tracking System**" carried out by Ms. **Vedashree K A** bearing USN **1JB24CS414** is bonafide students of **SJB Institute of Technology** in partial fulfilment for 6th semester of **BACHELOR OF ENGINEERING** in **Computer Science & Engineering** of the **Visvesvaraya Technological University**, **Belagavi** during the academic year **2025-26.** It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the Report deposited in the Departmental library. The project report has been approved as it satisfies the academic requirements in respect of Mini Project prescribed for the said Degree.

Signature of Class Coordinator

Signature of HOD
Dr. Krishna A N
Professor & Head
Dept. of CSE, SJBIT

ACKNOWLEDGEMENT

We would like to express our profound grateful to His Divine Soul **Jagadguru Padmabhushan Sri Sri Sri Dr. Balagangadharanatha Mahaswamiji** and His Holiness **Jagadguru Sri Sri Sri Dr. Nirmalanandanatha Mahaswamiji** for providing us an opportunity to complete our academics in this esteemed institution.

We would also like to express our profound thanks to **Revered Sri Sri Dr. Prakashnath Swamiji**, Managing Director, BGS & SJB Group of Institutions, for his continuous support in providing amenities to carry out this Mini Project Work in this admired institution.

We express our gratitude to **Dr. Puttaraju,** Academic Director, BGS & SJB Group of Institutions, for providing us an excellent facilities and academic ambience, which have helped us in satisfactory completion of Mini Project work.

We express our gratitude to **Dr. K. V. Mahendra Prashanth,** Principal, SJB Institute of Technology, for providing us an excellent facilities and academic ambience; which have helped us in satisfactory completion of Mini Project work.

We extend our heartfelt gratitude to all the Deans of SJB Institute of Technology for their unwavering support, cutting-edge facilities, and the inspiring academic environment, all of which played a pivotal role in the successful completion of our mini project work.

We extend our sincere thanks to **Dr. Krishan A N,** Head of the Department, Computer Science and Engineering, for providing us an invaluable support throughout the period of our Mini Project work.

We wish to express our heartfelt gratitude to our guide **Mr. Santhosh K**, Department of CSE for his valuable guidance, suggestions and cheerful encouragement during the entire period of our Mini Project work.

Finally, we take this opportunity to extend our earnest gratitude and respect to our parents, Teaching & Non teaching staffs of the department, the library staff and all our friends, who have directly or indirectly supported us during the period of our Mini Project work.

Regards

Vedashree K A

TABLE OF CONTENTS

CHAPTER NO	TOPIC	PAGE NO
1.	Introduction	1
2.	Problem statement	2
3.	Methodologies	3
4.	Machine Learning Approach	4
5.	Training Workflow	5
6.	System Architecture	6-7
7.	Implementation	8-10
8.	Testing and Result	11-15
9.	Conclusion	16

LIST OF FIGURES

FIGURE NO	DESCRIPTION	PAGE NO
Fig 6.1	System Architecture	6
Fig 8.1	Home page	11
Fig 8.2	Focused state output	11
Fig 8.3	Distracted state output	12
Fig 8.4	Drowsy state output	13
Fig 8.5	Teacher dashboard	13
Fig 8.6	Attention score over time	14

CHAPTER 1

INTRODUCTION

In recent years, the rapid growth of digital learning platforms and online education has transformed the traditional methods of teaching and learning. While these advancements have improved accessibility and flexibility, they have also introduced new challenges, particularly in maintaining student attention and engagement during learning sessions. In both physical classrooms and virtual environments, students are often prone to distractions caused by mobile devices, social media, fatigue, and lack of supervision. As a result, reduced attention spans can negatively impact learning outcomes and academic performance.

Conventional methods of monitoring student attention rely primarily on manual observation by teachers, which is subjective, time-consuming, and impractical in large classrooms or online settings. Moreover, in remote learning environments, it becomes even more difficult for instructors to assess whether students are actively engaged or distracted. This highlights the need for an automated, intelligent system capable of monitoring attention levels in real time without requiring human intervention.

Recent advancements in Artificial Intelligence (AI) and Computer Vision have opened new possibilities for analysing human behaviour through visual data. Technologies such as facial recognition, eye tracking, and head pose estimation enable the development of systems that can interpret user attention based on visual cues. By leveraging these techniques, it is possible to design a system that continuously observes a student's facial features and predicts their level of engagement.

This project proposes an AI-based Student Attention Tracking System that utilizes computer vision techniques to monitor and analyse student behaviour through a webcam. The system employs libraries such as OpenCV and Media Pipe to detect facial landmarks, track eye movements, and analyze head orientation in real time. Based on these inputs, the system classifies the user's attention state into categories such as Focused, Distracted, and Drowsy.

The primary goal of this system is to provide a non-intrusive, cost-effective, and scalable solution for attention monitoring. By offering real-time feedback, the system helps students become more aware of their focus levels and encourages better study habits. Additionally, this technology can be integrated into smart classrooms and e-learning platforms to enhance the overall learning experience.

CHAPTER 2

PROBLEM STATEMENT

Maintaining student attention during learning sessions is a critical challenge in both traditional classrooms and modern digital learning environments. With the increasing adoption of online education, it has become more difficult for instructors to monitor student engagement effectively. Students are often distracted by external factors such as mobile devices, social media, fatigue, and environmental disturbances, which negatively impact their concentration and academic performance.

Conventional methods of tracking attention rely on manual observation by teachers, which is subjective, inconsistent, and not scalable for large classrooms or remote learning platforms. In virtual settings, the lack of direct supervision further limits the ability to assess whether students are attentive or disengaged. As a result, there is no reliable and automated mechanism to continuously evaluate student focus in real time.

Existing technological solutions are either complex, expensive, or require specialized hardware, making them impractical for widespread use in educational institutions. Additionally, many systems fail to provide real-time feedback or actionable insights that can help improve student learning behavior.

Therefore, there is a need for a cost-effective, non-intrusive, and intelligent system that can automatically monitor and analyze student attention using readily available devices such as webcams. The system should be capable of detecting behavioral indicators such as eye movement, facial orientation, and drowsiness, and classify attention levels accurately in real time. Such a solution can significantly enhance the learning experience by enabling better engagement tracking and timely intervention.

CHAPTER 3

METHODOLOGIES

3.1 Overview

The proposed system, *FocusAI: Real-Time Student Attention Tracking System*, is designed to monitor and evaluate student attention using computer vision techniques. It follows a structured pipeline that processes real-time video input from a webcam to analyze facial features such as eye movement and head orientation. Based on this analysis, the system determines whether the user is focused, distracted, or drowsy. This approach ensures continuous monitoring and real-time feedback.

3.2 System Architecture

The system architecture consists of multiple sequential stages, including webcam input, frame capture, face detection, landmark extraction, eye and head analysis, attention classification, and output display. Each module performs a specific function, ensuring smooth data flow and efficient processing.

3.3 Video Acquisition and Face Detection

The system captures live video using a webcam through OpenCV, where frames are continuously processed. MediaPipe's Face Mesh model is used to detect faces and extract facial landmarks. These landmarks represent key facial features such as eyes, nose, and mouth, providing essential spatial information for further analysis.

3.4 Eye Tracking and Head Pose Analysis

Eye tracking is used to analyze blinking patterns and eye closure to detect drowsiness. If the eyes remain closed for a prolonged duration, the user is classified as drowsy. Head pose estimation determines the direction of the user's face. If the head is facing forward, the user is considered focused, These combined analyses help in accurately understanding user attention.

3.5 Attention Classification and Output

The system uses a rule-based approach to classify attention into focused, distracted, and drowsy states based on eye and head behavior. The results are displayed in real time on the video feed using OpenCV, with color indicators such as green for focused, red for distracted, and yellow for drowsy. The methodology is efficient, non-intrusive, and easy to implement, though it may be affected by lighting conditions and camera quality.

CHAPTER 4
MACHINE LEARNING APPROACH

The proposed system primarily follows a computer vision–based pipeline supported by a lightweight machine learning approach to enhance attention detection accuracy. While the core implementation relies on facial landmark extraction using pre-trained models, machine learning concepts are incorporated to improve the classification of attention states based on behavioral patterns.

The system utilizes a pre-trained facial landmark detection model provided by MediaPipe, which is trained on large-scale datasets to accurately identify facial features. This model eliminates the need for training from scratch and enables efficient real-time detection of key facial points such as eyes, nose, and mouth. These landmarks serve as input features for further analysis. Feature extraction plays a crucial role in the machine learning approach. Important features such as eye aspect ratio, blink frequency, and head orientation are derived from the detected landmarks. These features represent the behavioral characteristics of the user and are used to determine attention levels. For example, a decrease in eye aspect ratio indicates eye closure, while significant variation in head orientation suggests distraction.

The system adopts a rule-based classification mechanism as a lightweight alternative to complex machine learning models. However, this approach can be extended by integrating supervised learning algorithms such as Decision Trees, Support Vector Machines (SVM), or Random Forest classifiers. In such cases, the extracted features can be used to train a model on labeled datasets containing different attention states. The trained model can then predict the user's attention level more accurately under varying conditions.

To maintain real-time performance, the current implementation avoids computationally intensive deep learning models. Instead, it focuses on efficient feature extraction and logical decision-making. This ensures that the system can run smoothly on standard computing devices without requiring high-end hardware.

In future enhancements, deep learning techniques such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) can be incorporated to improve accuracy and robustness. These models can learn complex patterns from large datasets and provide more reliable attention predictions In summary, the machine learning approach in this system combines pre-trained models, feature extraction, and rule-based classification to achieve a balance between accuracy and efficiency. This approach makes the system practical, scalable, and suitable for real-time educational applications.

CHAPTER 5

TRAINING WORKFLOW

The testing workflow of the proposed system focuses on evaluating the performance and reliability of the attention tracking model in real-time conditions. This phase ensures that the system accurately detects and classifies user attention states such as focused, distracted, and drowsy using live webcam input.

The testing process begins with real-time video capture, where the webcam continuously streams frames. Each frame is processed individually using OpenCV to maintain smooth and efficient execution. The captured frames are then passed to the facial detection module, where MediaPipe Face Mesh identifies the presence of a face and extracts facial landmarks.

Once the landmarks are obtained, feature extraction is performed to derive meaningful parameters such as eye aspect ratio, blink patterns, and head orientation. These features are essential for determining the user's attention level. The extracted features are then passed to the classification module.

In the current implementation, a rule-based approach is used for testing, where predefined conditions are applied to classify the attention state. If a trained machine learning model is integrated, the extracted features are fed into the model, which predicts the attention class based on learned patterns.

The system continuously evaluates each frame and updates the attention status in real time. The output is displayed on the screen along with visual indicators, ensuring immediate feedback. During testing, various conditions such as lighting variations, different head movements, and multiple user behaviors are considered to evaluate system robustness.

Performance evaluation is carried out by observing the accuracy of attention classification under different scenarios. Metrics such as accuracy and response time are analyzed to ensure that the system performs efficiently in real-time environments.

In summary, the testing workflow validates the effectiveness of the system by processing live video input, extracting features, classifying attention states, and displaying results. This ensures that the system is reliable, responsive, and suitable for practical use in educational environments.

CHAPTER 6

SYSTEM ARCHITECTURE

The system architecture of the proposed student attention tracking system is designed as a sequential pipeline that processes real-time video input and analyzes user behavior to determine attention levels. The architecture consists of multiple interconnected modules, each responsible for a specific task such as video capture, face detection, feature extraction, and attention classification. This modular design ensures efficient data flow, real-time processing, and

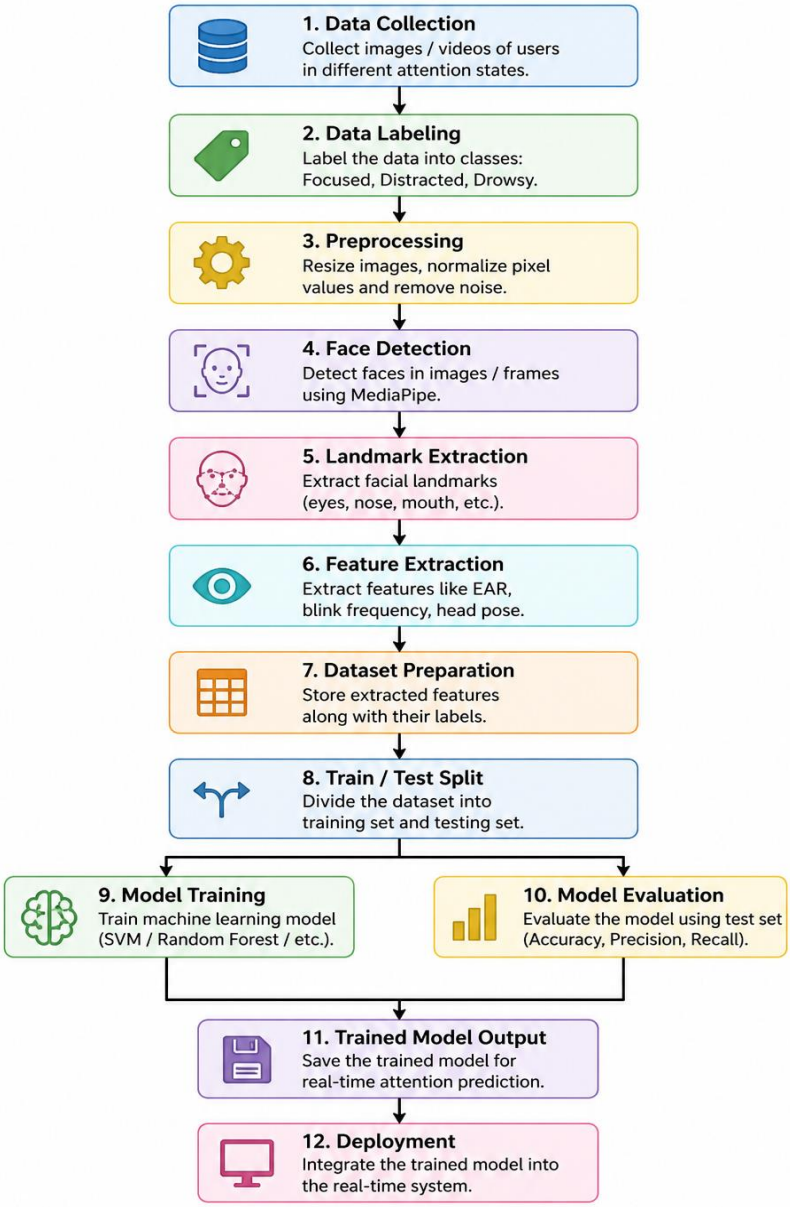

Fig 6.1: System Architecture

As shown in the above diagram, the system begins with the webcam input, which captures real-time video of the user. The captured video is divided into individual frames using the frame capture module. These frames are then passed to the face detection module, where the presence of a face is identified using computer vision techniques.

Once a face is detected, the landmark extraction module identifies key facial points such as the eyes, nose, and mouth. These landmarks are essential for analyzing facial behavior. The extracted landmarks are then processed in the eye and head analysis module, where eye movement, blinking patterns, and head orientation are evaluated.

Based on this analysis, the attention classification module determines the user's state as focused, distracted, or drowsy using predefined logic or a trained model. Finally, the output display module presents the result in real time by overlaying the attention status on the video feed.

This architecture ensures a smooth and efficient workflow, enabling accurate real-time attention monitoring with minimal computational overhead.

CHAPTER 7

IMPLEMENTATION

7.1 Overview

The proposed system, *FocusAI: Real-Time Student Attention Tracking System*, is implemented as a modular, real-time web application using Python. The architecture is divided into several interconnected subsystems, including Video Acquisition, Facial Feature Extraction, Behavioral Analysis, and User Interface. The system is designed to process live webcam input and provide real-time attention analysis with efficient performance and scalability.

7.2 System Architecture and Technology Stack

The system is developed using a modern Python-based technology stack. The frontend and dashboard are implemented using **Streamlit**, which enables the creation of an interactive user interface and teacher analytics dashboard. Computer vision operations are performed using **OpenCV**, which handles webcam integration and frame processing. Facial landmark detection is achieved using **MediaPipe Face Mesh**, which provides 468 real-time 3D facial landmarks.

For data processing and visualization, libraries such as **NumPy**, **Pandas**, and **Plotly Express** are utilized. The system uses **SQLite** for storing attention tracking data.

7.3 Core Modules

The implementation is structured into multiple modules to ensure maintainability and scalability.

7.3.1 Video Acquisition and Face Detection

The system captures live video using OpenCV's VideoCapture functionality. Each frame is processed using MediaPipe Face Mesh, which extracts 468 facial landmarks in three-dimensional space. These landmarks serve as the primary input for further analysis.

7.3.2 Eye Tracking and Drowsiness Detection

Eye tracking is performed using the Eye Aspect Ratio (EAR), which measures the distance between vertical and horizontal eye landmarks. If the EAR falls below a predefined threshold (e.g., 0.22) for a continuous duration, the system detects eye closure and classifies the user as drowsy.

Additionally, Mouth Aspect Ratio (MAR) is used to detect yawning by analyzing the distance between upper and lower lip landmarks. This acts as a secondary indicator of drowsiness.

7.3.3 Head Pose Estimation

Head pose estimation is implemented using the Perspective-n-Point (SolvePnP) algorithm. By mapping 2D facial landmark points to a generic 3D face model, the system calculates head rotation angles such as yaw, pitch, and roll.

Based on these angles, the system determines whether the user is looking forward or away (left, right, up, or down), enabling accurate distraction detection.

7.3.4 Attention Logic and Scoring

The extracted behavioral features are used to classify attention into three categories. When the user's eyes are open and the head is facing forward, the system classifies the state as focused and assigns a score of 100%. If the head is turned away while the eyes remain open, the state is classified as distracted with a score of 40%. If eye closure or yawning is detected, the system classifies the state as drowsy with a score of 0%.

This rule-based scoring mechanism ensures real-time performance and simplicity.

7.4 Real-Time Alerts and Database Logging

To avoid false detections caused by brief movements such as blinking, the system implements a timer-based mechanism. If the user remains in a distracted or drowsy state for more than five seconds, the system triggers a visual alert and plays an audio notification to regain attention.

The system also integrates a database module that logs attention data periodically. Each record includes the timestamp, attention status, and score, which are stored in a local SQLite database for further analysis.

7.5 Interactive Dashboard

The application interface is developed using Streamlit and consists of two main components. The Live Tracking interface displays the real-time video feed with facial landmarks and attention status indicators.

It includes dynamic color coding, where green represents focused, red indicates drowsy, and yellow indicates distraction.

The Teacher Analytics Dashboard provides insights into historical attention data by querying the SQLite database. It computes performance metrics such as average attention score and visualizes engagement trends using interactive charts, enabling educators to analyze student behavior over time.

7.6 Summary

The implementation integrates computer vision, machine learning models, and interactive visualization to create a robust attention tracking system. The modular architecture ensures scalability, while the use of efficient libraries enables real-time performance. The system effectively combines feature extraction, behavioral analysis, and user interface design to deliver accurate and practical attention monitoring.

CHAPTER 8

TESTING AND RESULT

Testing is a crucial phase in the development of the FocusAI: Real-Time Student Attention Tracking System. It ensures that the system performs accurately, efficiently, and reliably under different real-time conditions. The main objective of testing is to evaluate whether the system can correctly identify various attention states such as focused, distracted, and drowsy using computer vision techniques.

The system was tested using a webcam in a real-time environment with different user behaviors. Various conditions such as normal attention, head movement, and eye closure were simulated to analyze system performance. The results obtained from testing confirm that the system effectively detects user attention and provides accurate real-time feedback.

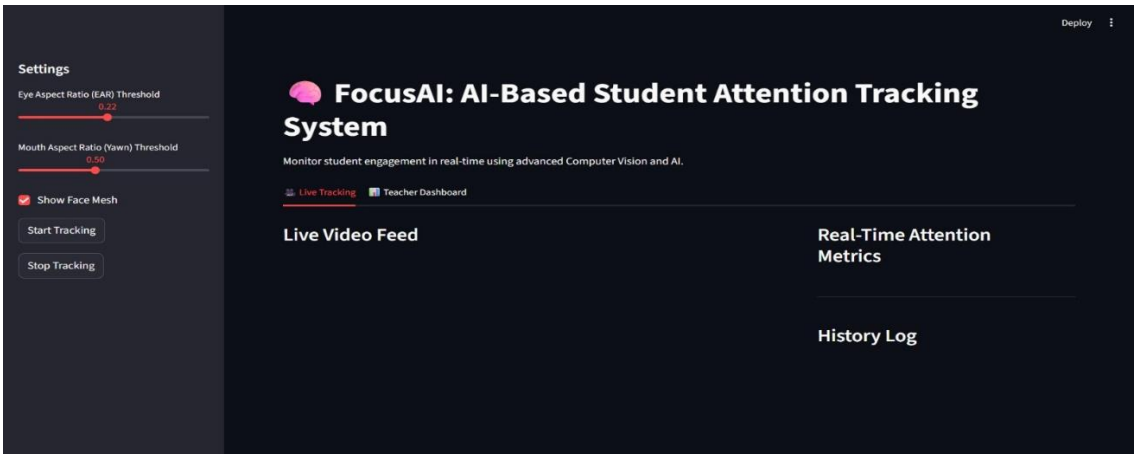

Fig 8.1: Home page

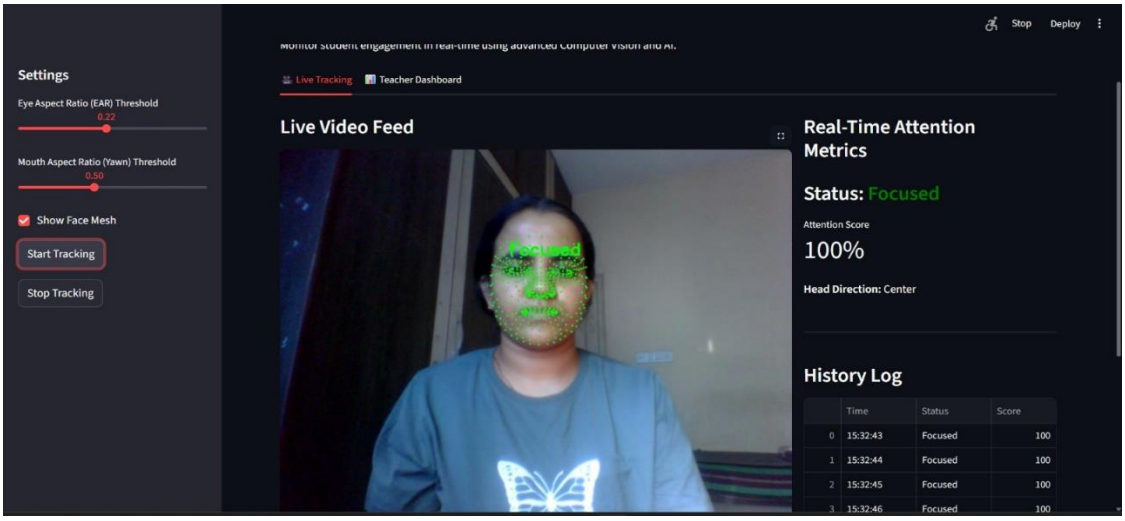

Fig 8.2: Focused State Output

The above figure shows the system detecting a focused state. In this condition, the user is looking directly at the screen with eyes open and head positioned at the center. The MediaPipe Face Mesh model accurately identifies facial landmarks, and the system calculates stable eye and head parameters.

Since both eye openness and head direction meet the required conditions, the system classifies the user as **"Focused"**. The attention score is displayed as **100%**, indicating maximum engagement. This demonstrates that the system can reliably identify attentive behavior in real tim

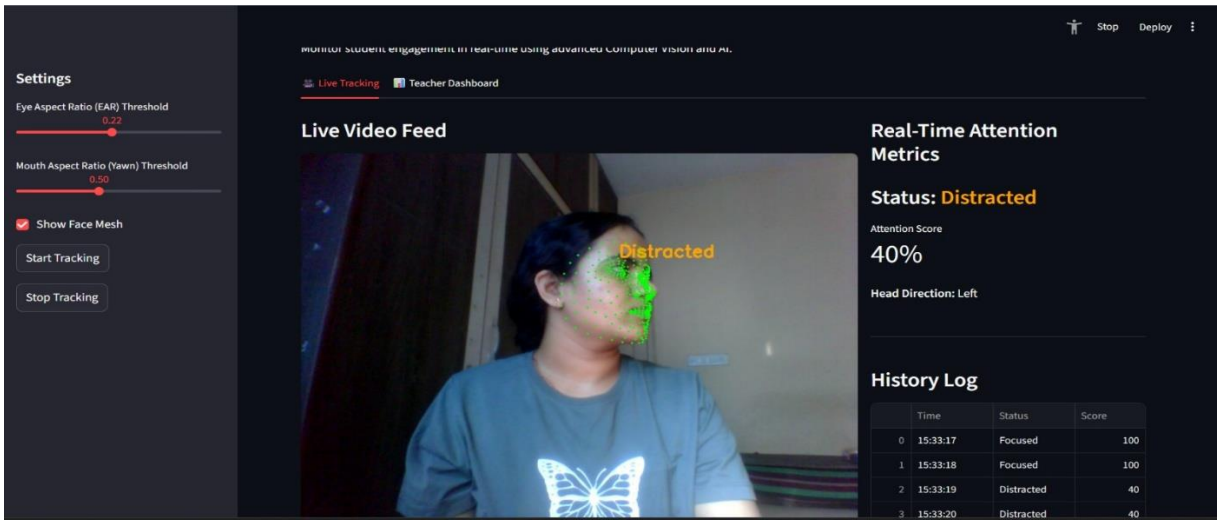

Fig 8.3: Distracted State Output

The above figure represents the distracted condition. In this scenario, the user turns their head away from the screen. The head pose estimation algorithm detects this deviation and identifies that the user is not looking at the screen.

Although the eyes may still be open, the system prioritizes head direction for attention analysis. As a result, the system classifies the user as **"Distracted"** and reduces the attention score to around **40%**. This confirms the system's ability to detect loss of focus accurately.

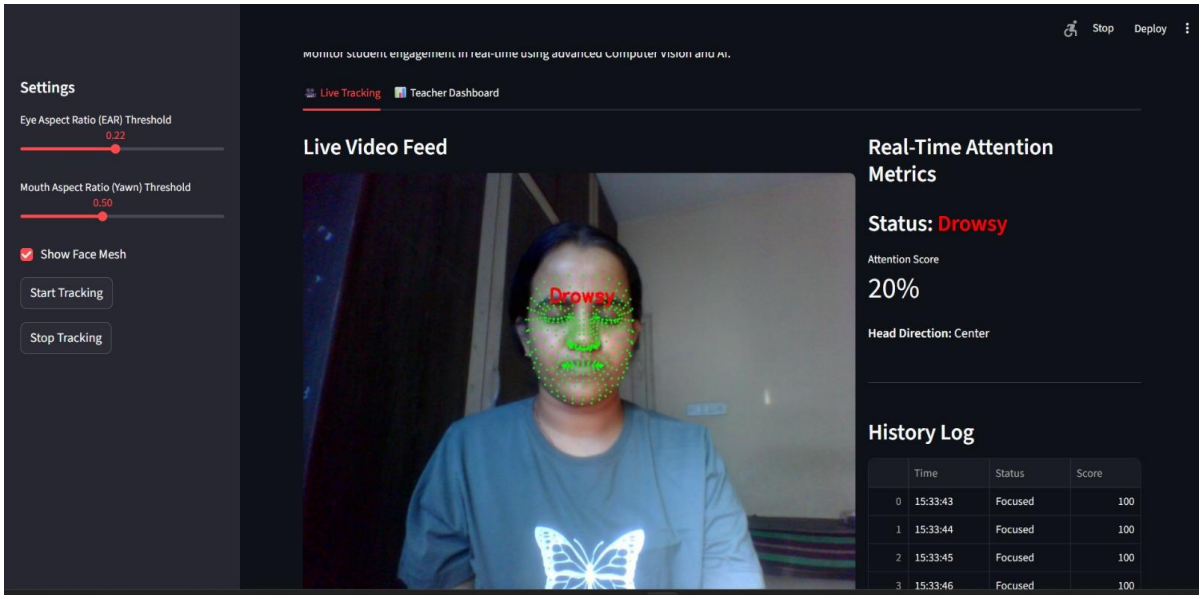

Fig 8.4: Drowsy State Output

The above figure shows the system detecting a drowsy state. In this case, the user's eyes are partially or completely closed. The Eye Aspect Ratio (EAR) falls below the defined threshold, indicating reduced alertness.

The system identifies this behavior as drowsiness and classifies the user as **"Drowsy"**. The attention score drops significantly (around **20% or lower**), indicating low engagement. This proves that the system can effectively detect fatigue and reduced attention levels.

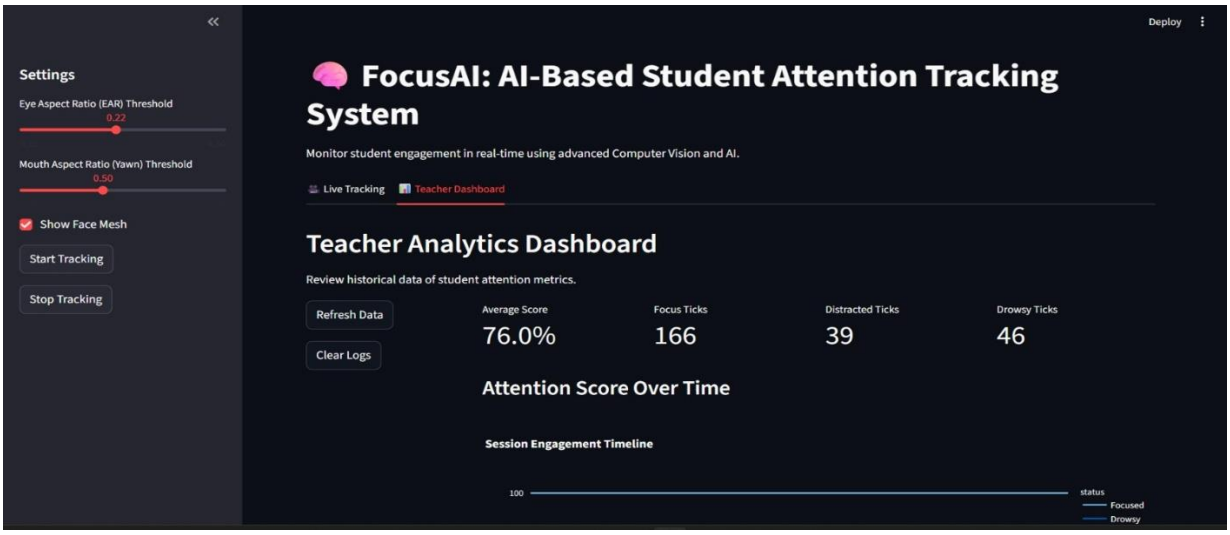

Fig 8.5: Teacher Dashboard

The above figure illustrates the teacher analytics dashboard, which provides a comprehensive overview of student attention data. The dashboard displays key performance indicators such as average attention score, number of focused instances, distracted counts, and drowsy occurrences.

This feature enables educators to monitor student engagement over time. The dashboard also provides a clear and structured view of performance, helping teachers identify patterns and take corrective actions if necessary.

Figure 8.6: Attention Score Over Time

The above graph represents the variation of attention score over time. It visually demonstrates how the user's attention fluctuates between focused, distracted, and drowsy states.

This graphical representation helps in understanding behavioral trends and identifying specific time intervals where the user lost focus. It enhances data interpretation and provides valuable insights into attention patterns.

The above table shows the recorded data stored in the system database. Each entry contains details such as timestamp, attention status, and attention score.

The use of SQLite ensures efficient data storage and retrieval. This data can be used for further analysis, report generation, and long-term tracking of student performance.

Results and Performance Analysis

The testing results clearly indicate that the system performs efficiently in real-time conditions. It successfully identifies all three attention states with good accuracy. The integration of facial landmark detection, eye tracking, and head pose estimation ensures reliable classification.

The system operates smoothly without requiring high computational power, making it suitable for practical applications in educational environments. The real-time feedback mechanism also enhances user awareness and helps improve focus.

Conclusion of Testing

The FocusAI system has been successfully tested and validated across multiple scenarios. The results confirm that the system is capable of accurately detecting attention levels and providing meaningful insights.

Overall, the system proves to be reliable, scalable, and effective for monitoring student engagement in online learning environments. It bridges the gap between traditional classroom monitoring and digital education, making it a valuable tool for both students and educators.

CHAPTER 9

CONCLUSION

The proposed system, *FocusAI: Real-Time Student Attention Tracking System*, successfully demonstrates the practical application of Computer Vision and Artificial Intelligence in the educational technology domain. By integrating advanced libraries such as OpenCV and MediaPipe Face Mesh, the system effectively automates the otherwise subjective and challenging task of monitoring student engagement during digital learning sessions.

The system accurately extracts 3D facial landmarks and computes essential behavioral metrics, including Eye Aspect Ratio (EAR) for detecting drowsiness and Head Pose Estimation for identifying distraction. Through real-time processing, the system classifies user attention into three distinct states: Focused, Distracted, and Drowsy. The implementation ensures high accuracy and efficiency without requiring high-end computational resources, making it suitable for deployment on standard devices.

Furthermore, the integration of a web-based dashboard using Streamlit, along with SQLite database support, enhances the system's functionality by providing both real-time feedback and historical analytics. The system not only alerts users through visual and auditory notifications but also stores attention data for further analysis, enabling educators to monitor engagement trends effectively.

Overall, the FocusAI system bridges the gap between traditional classroom monitoring and modern remote learning environments. It offers a scalable, non-intrusive, and intelligent solution for improving student focus and academic productivity. The project successfully meets its objectives and provides a strong foundation for future advancements in real-time attention tracking systems.